

Physics Notes

Lecture 1: Units, Measurement Systems, and Vectors

Lecture Notes and Worked Examples

These notes develop the material covered in our first lecture, following Young and Freedman, University Physics with Modern Physics, 13th edition, Chapter 1, with the additional unit systems and exercises discussed in class.

Green boxes give definitions, blue boxes collect rules, purple boxes explain derivations, red boxes contain worked examples, and orange boxes flag common mistakes. The supplied textbook Figures 1.27 and 1.29 are included and explained. Practice problems are collected at the end of the notes.

Contents

1	Units, Measurement Systems, and Vectors	1
1.1	Physical Quantities and Measurement	1
1.2	SI Base Quantities and Fundamental Units	2
1.3	Derived Quantities, Units, and Dimensional Checks	2
1.4	The CGS System	3
1.5	Multiples, Submultiples, and Unit Conversions	4
1.6	Scalars, Vectors, and Magnitudes	5
1.7	Vector Addition, Subtraction, and Scalar Multiplication	6
1.8	Components, Unit Vectors, and Direction Angles	7
1.9	Worked Displacement Example: The Buried Keys	8
1.10	The Scalar or Dot Product	9
1.11	The Vector or Cross Product	12
	Quick Reference and Common Exam Traps	15
	Practice Problems	16

1 Units, Measurement Systems, and Vectors

Notation. Bold symbols such as $\mathbf{A}$ denote vectors. The corresponding plain symbol $A = |\mathbf{A}|$ denotes a nonnegative magnitude. Unless a question says otherwise, use a right-handed Cartesian coordinate system, measure planar direction angles counterclockwise from $+x$, and set the calculator to **degree mode** for angles written in degrees.

1.1 Physical Quantities and Measurement

Young–Freedman, Sec. 1.3–1.4, pp. 4–7.

Definition: A physical quantity needs a number and a unit

A physical quantity is something that can be measured or calculated from measurements. Write it as

$$\text{quantity} = \text{numerical value} \times \text{unit}.$$

For example, a length $\ell = 1.25 \text{ m}$ has numerical value 1.25 when its unit is the metre. The *same length* is 125 cm: changing the unit changes the number, not the physical quantity.

Quantity	Example from the lecture	Meaning of the unit
Length	2.5 m	metre
Mass	12 kg	kilogram
Time interval	3 s	second
Temperature	300 K	kelvin
Force magnitude	15 N	newton
Energy	50 J	joule
Pressure	100 kPa	kilopascal

Example: Direct and indirect measurements

Direct: read a length from a ruler, for example 1.25 m.

Indirect: measure other quantities and use a relation. If an object travels a distance $d = 100 \text{ m}$ in $\Delta t = 5 \text{ s}$, its *average speed* is

$$v_{\text{av}} = \frac{d}{\Delta t} = \frac{100 \text{ m}}{5 \text{ s}} = 20 \text{ m s}^{-1}.$$

This quotient does not prove that its speed was constant during the journey. Likewise, a sample's average density is

$$\rho = \frac{m}{V},$$

where m is mass and V is volume. Common density units are kg m^{-3} and g cm^{-3} .

Watch out

Write the unit at every step. “The length is 1.25” is incomplete unless a unit has already been specified. Unit symbols are case-sensitive: m is metre, M is the prefix mega, and N is newton. Use a space between a number and its unit, as in 12 kg. Kelvin is written K, without a degree sign.

1.2 SI Base Quantities and Fundamental Units**Definition: The seven SI base quantities**

SI is the International System of Units. Its seven base quantities and units are:

Base quantity	SI base unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

The *quantity* and its *unit* are different things: mass is a quantity; kilogram is its SI base unit. In mechanics, the base units used most often are metre, kilogram, and second.

Watch out

The SI base unit of mass is the **kilogram**, not the gram. Mass and weight are different quantities: mass is measured in kilograms, while weight is a gravitational force measured in newtons. The mole measures amount of substance, not mass; converting moles into grams requires the substance’s molar mass.

1.3 Derived Quantities, Units, and Dimensional Checks

Young–Freedman, Sec. 1.3–1.4, pp. 4–7.

Definition: Derived units are combinations of base units

Quantity	Relation or example	SI unit
Area	length $\times$ width	m^2
Volume	length $\times$ width $\times$ height	m^3
Speed	distance/time	m s^{-1}
Acceleration	velocity change/time	m s^{-2}
Density	$\rho = m/V$	kg m^{-3}
Force	$\mathbf{F}_{\text{net}} = m\mathbf{a}$	$\text{N} = \text{kg m s}^{-2}$
Work or energy	$W = \mathbf{F} \cdot \mathbf{d}$	$\text{J} = \text{kg m}^2 \text{s}^{-2}$
Pressure	$P = F_{\perp}/\mathcal{A}$	$\text{Pa} = \text{kg m}^{-1} \text{s}^{-2}$

Here $\mathcal{A}$ denotes the area on which a normal force $F_{\perp}$ acts. N, J, and Pa are special names for derived units, not additional base units.

Derivation: Where the newton, joule, and pascal come from

From Newton's second law, force has units mass $\times$ acceleration:

$$1 \text{ N} = 1 \text{ kg m s}^{-2}.$$

For a constant force parallel to a displacement, $W = Fd$. Hence

$$1 \text{ J} = 1 \text{ N m} = 1 \text{ kg m}^2 \text{ s}^{-2}.$$

If the force is not parallel to the displacement, use $W = Fd \cos \phi$, as explained in Section 1.10. Dividing a normal force by an area gives

$$1 \text{ Pa} = 1 \text{ N m}^{-2} = 1 \text{ kg m}^{-1} \text{ s}^{-2}.$$

Therefore $100 \text{ kPa} = 100 \times 10^3 \text{ Pa} = 10^5 \text{ Pa}$.

Theorem / Rule: Use dimensions to detect impossible equations

The *dimension* of a quantity describes its physical type, independently of the chosen unit. Write $[\ell] = L$, $[m] = M$, $[t] = T$. Then

$$[v] = LT^{-1}, \quad [a] = LT^{-2}, \quad [\rho] = ML^{-3}.$$

Two sides of an equation must have the same dimensions, and quantities can be added only if they have compatible dimensions. For example, $d = vt$ is dimensionally possible because $(LT^{-1})T = L$, but $d = v + t$ is not.

Dimensional agreement is a *necessary check*, not a proof of correctness. It cannot distinguish $d = vt$ from $d = 2vt$, since the number 2 has no dimensions. Nor does it allow adding unrelated quantities merely because their units match.

1.4 The CGS System

Definition: Centimetre–gram–second

CGS uses the centimetre, gram, and second as its basic mechanical units:

$$1 \text{ cm} = 10^{-2} \text{ m}, \quad 1 \text{ g} = 10^{-3} \text{ kg}, \quad 1 \text{ s} = 1 \text{ s}.$$

The CGS unit of force is the **dyne**; the CGS unit of energy is the **erg**.

Derivation: Convert CGS force and energy into SI

Apply $F = ma$ using CGS units:

$$\begin{aligned} 1 \text{ dyne} &= 1 \text{ g cm s}^{-2} \\ &= (10^{-3} \text{ kg})(10^{-2} \text{ m}) \text{ s}^{-2} = 10^{-5} \text{ N}. \end{aligned}$$

Therefore

$$1 \text{ dyne} = 10^{-5} \text{ N}, \quad 1 \text{ N} = 10^5 \text{ dyne}.$$

Energy is force $\times$ distance, so

$$1 \text{ erg} = 1 \text{ dyne cm} = (10^{-5} \text{ N})(10^{-2} \text{ m}) = 10^{-7} \text{ J}.$$

Thus

$$1 \text{ erg} = 10^{-7} \text{ J}, \quad 1 \text{ J} = 10^7 \text{ erg}.$$

For example, $2.0 \text{ N} = 2.0 \times 10^5 \text{ dyne}$, and $3.5 \times 10^6 \text{ erg} = 0.35 \text{ J}$.

1.5 Multiples, Submultiples, and Unit Conversions**Theorem / Rule: Prefixes used in this lecture**

A prefix multiplies the unit immediately following it by a power of ten. This is a useful working list, not the complete SI prefix list.

Prefix	Symbol	Factor	Prefix	Symbol	Factor
tera	T	10^{12}	deci	d	10^{-1}
giga	G	10^9	centi	c	10^{-2}
mega	M	10^6	milli	m	10^{-3}
kilo	k	10^3	micro	μ	10^{-6}
hecto	h	10^2	nano	n	10^{-9}
deca	da	10^1	pico	p	10^{-12}
			femto	f	10^{-15}

Milli is needed for both millimetres and milligrams. For example,

$$1 \text{ cm} = 10 \text{ mm}, \quad 1 \text{ g} = 1000 \text{ mg}, \quad 1 \text{ mg} = 10^{-6} \text{ kg}.$$

In mm, the first m is the prefix milli and the second m is the unit metre.

Theorem / Rule: Conversion factors are equal to one

Multiply by a ratio of equal physical quantities, with the unwanted unit in the denominator:

$$2.5 \text{ km} \times \frac{1000 \text{ m}}{1 \text{ km}} = 2500 \text{ m}.$$

For a rate, convert numerator and denominator:

$$72 \text{ km h}^{-1} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \boxed{20 \text{ m s}^{-1}}.$$

The units cancel algebraically. For these units, dividing a speed in km/h by 3.6 converts it to m/s; multiplying by 3.6 converts back.

Example: Squared units, cubed units, and density

Area: square the entire conversion factor:

$$1 \text{ cm}^2 = (10^{-2} \text{ m})^2 = 10^{-4} \text{ m}^2.$$

So $25 \text{ cm}^2 = 2.5 \times 10^{-3} \text{ m}^2$.

Volume: cube the factor:

$$1 \text{ cm}^3 = (10^{-2} \text{ m})^3 = 10^{-6} \text{ m}^3.$$

Useful volume relations are $1 \text{ mL} = 1 \text{ cm}^3$ and $1 \text{ L} = 10^{-3} \text{ m}^3$.

Density: convert mass and volume separately:

$$1 \text{ g cm}^{-3} = \frac{10^{-3} \text{ kg}}{10^{-6} \text{ m}^3} = 10^3 \text{ kg m}^{-3}.$$

A sample of mass 54 g and volume 20 cm^3 has density

$$\rho = \frac{54}{20} \text{ g cm}^{-3} = 2.7 \text{ g cm}^{-3} = 2700 \text{ kg m}^{-3}.$$

Watch out

$1 \text{ cm}^2 \neq 10^{-2} \text{ m}^2$: a prefix inside a squared or cubed unit must also be squared or cubed. Similarly, 1 mg is 10^{-3} g but 10^{-6} kg . Write the conversion instead of relying on a memorized direction for moving the decimal point.

Rounding habit. Keep extra digits in intermediate calculations and round the final answer. Exact conversion factors do not reduce the precision of a measurement. In the measured vector examples below, intermediate components are shown with extra digits; final results are rounded appropriately. Do not interpret all displayed calculator digits as measured precision.

1.6 Scalars, Vectors, and Magnitudes

Young–Freedman, Sec. 1.7–1.9, pp. 11–19.

Definition: Scalar versus vector

A **scalar** is specified by a numerical value and, when applicable, a unit. Examples include mass, time, temperature, energy, density, distance, and speed.

A **vector** has a magnitude and a spatial direction. Examples include displacement, velocity, acceleration, and force. Its components depend on the chosen coordinate axes. For perpendicular Cartesian axes,

$$\mathbf{v} = (v_x, v_y) \quad \text{in 2D}, \quad \mathbf{v} = (v_x, v_y, v_z) \quad \text{in 3D}.$$

Each component is a signed scalar. Components of a physical vector have the same units as that vector.

Derivation: Magnitude from perpendicular components

Pythagoras' theorem gives

$$|\mathbf{v}| = \sqrt{v_x^2 + v_y^2} \quad (2\text{D}), \quad |\mathbf{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2} \quad (3\text{D}).$$

For the lecture example $\mathbf{v} = (3, 5)$,

$$|\mathbf{v}| = \sqrt{3^2 + 5^2} = \sqrt{34} \approx 5.83.$$

If instead $\mathbf{v} = (3\hat{i} + 5\hat{j}) \text{ m s}^{-1}$, its magnitude is $\sqrt{34} \text{ m s}^{-1}$. A magnitude cannot be negative even when a component is negative. The zero vector has zero magnitude and no defined direction.

Example: Distance and displacement are different

Walk 3 m east and then 3 m west. The total distance travelled is 6 m, while the net displacement is the zero vector. Average speed uses total distance divided by elapsed time; average velocity uses displacement divided by elapsed time.

1.7 Vector Addition, Subtraction, and Scalar Multiplication

Young–Freedman, Sec. 1.7, pp. 11–14.

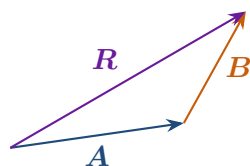
Theorem / Rule: Two graphical constructions for the same sum

For $\mathbf{R} = \mathbf{A} + \mathbf{B}$:

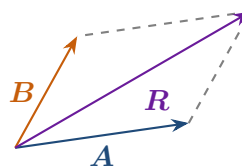
1. **Head-to-tail:** draw $\mathbf{A}$, then translate $\mathbf{B}$ so its tail is at the head of $\mathbf{A}$. Draw $\mathbf{R}$ from the first tail to the final head.
2. **Parallelogram:** draw both vectors from a common tail, complete the parallelogram, and draw its diagonal from the common tail to the opposite corner.

Translating a vector for a construction does not change its magnitude or direction. Addition is commutative and associative:

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}, \quad (\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}).$$



Head-to-tail



Parallelogram

Definition: Subtract by adding the negative vector

The vector $-\mathbf{B}$ has the same magnitude as $\mathbf{B}$ and the opposite direction. Therefore

$$\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B}).$$

If $\mathbf{A}$ and $\mathbf{B}$ share a tail, the arrow *from the head of $\mathbf{B}$ to the head of $\mathbf{A}$* represents $\mathbf{A} - \mathbf{B}$. Order matters: $\mathbf{B} - \mathbf{A} = -(\mathbf{A} - \mathbf{B})$.

Theorem / Rule: Calculate component by component

$$(A_x, A_y) + (B_x, B_y) = (A_x + B_x, A_y + B_y),$$

$$(A_x, A_y) - (B_x, B_y) = (A_x - B_x, A_y - B_y).$$

In 3D, perform the same operation on the z -components. For a real number c ,

$$c\mathbf{A} = (cA_x, cA_y, cA_z), \quad |c\mathbf{A}| = |c| |\mathbf{A}|.$$

A positive c preserves direction, a negative c reverses it, and $c = 0$ produces the zero vector. For $\mathbf{A} = (5, 3)$, $c\mathbf{A} = (5c, 3c)$; in particular, $-2\mathbf{A} = (-10, -6)$.

Watch out

Do not add vector magnitudes unless the directions justify it. In general, $|\mathbf{A} + \mathbf{B}| \neq A + B$. For perpendicular vectors, $|\mathbf{A} + \mathbf{B}| = \sqrt{A^2 + B^2}$; for parallel vectors in the same direction, it is $A + B$; for opposite directions, it is $|A - B|$.

1.8 Components, Unit Vectors, and Direction Angles

Young–Freedman, Sec. 1.8–1.9, pp. 14–19.

Derivation: Resolving a vector into components

Let a nonzero vector $\mathbf{A}$ have magnitude A and direction angle θ measured counterclockwise from $+x$. The horizontal and vertical projections give

$$\cos \theta = \frac{A_x}{A}, \quad \sin \theta = \frac{A_y}{A}, \quad \boxed{A_x = A \cos \theta, \quad A_y = A \sin \theta.}$$

These formulas hold in every quadrant when signed sine and cosine values are used. The angle's reference axis is crucial: if the angle is measured from $+y$, the component adjacent to that angle is the y -component.

Definition: Unit-vector notation

The Cartesian unit vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ point along $+x$, $+y$, and $+z$. Each has magnitude one and no physical unit. Write

$$\boxed{\mathbf{A} = A_x \hat{\mathbf{i}} + A_y \hat{\mathbf{j}} + A_z \hat{\mathbf{k}}.}$$

For a 2D vector, omit the z -term. The coefficients carry the physical units. For any nonzero vector, its own direction unit vector is

$$\hat{\mathbf{A}} = \frac{\mathbf{A}}{|\mathbf{A}|}.$$

For example, $(3\hat{\mathbf{i}} + 4\hat{\mathbf{j}})/5$ is a unit vector, since its magnitude is one.

Theorem / Rule: Recover magnitude and direction from components

For a planar resultant,

$$R_x = A_x + B_x, \quad R_y = A_y + B_y, \quad R = \sqrt{R_x^2 + R_y^2}.$$

If $R_x \neq 0$, then $\tan \theta = R_y/R_x$, but an ordinary inverse tangent does not identify the quadrant by itself. First inspect the signs:

Quadrant	R_x	R_y	Standard angle
I	+	+	$0^\circ < \theta < 90^\circ$
II	-	+	$90^\circ < \theta < 180^\circ$
III	-	-	$180^\circ < \theta < 270^\circ$
IV	+	-	$270^\circ < \theta < 360^\circ$

Find the acute reference angle $\beta = \tan^{-1}(|R_y/R_x|)$, then choose $\theta = \beta$, $180^\circ - \beta$, $180^\circ + \beta$, or $360^\circ - \beta$, respectively. On an axis, read the direction directly; the zero vector has no direction.

Example: Compass directions: read the words before choosing sine or cosine

Take east as $+x$ and north as $+y$.

- **32° east of north:** start facing north and turn toward east. Then $A_x = A \sin 32^\circ$, $A_y = A \cos 32^\circ$. The standard angle is 58° .
- **36° south of west:** start facing west and turn toward south. Then $B_x = -B \cos 36^\circ$, $B_y = -B \sin 36^\circ$. The standard angle is 216° .

Both methods agree with $A_x = A \cos \theta$, $A_y = A \sin \theta$ when θ is the standard angle. A quick sketch fixes the signs before any calculation.

1.9 Worked Displacement Example: The Buried Keys

Young–Freedman, Sec. 1.8, Example 1.7, pp. 18.

Example: Complete the three-displacement calculation

The given displacements are

A : 72.4 m, 32.0° east of north,

B : 57.3 m, 36.0° south of west,

C : 17.8 m due south.

1. Set up. Place the origin at the starting point, with east as $+x$ and north as $+y$. We need $R = A + B + C$.

2. Resolve each displacement. Use standard angles 58.0° , 216.0° , and 270.0° :

$$\begin{aligned}A_x &= (72.4 \text{ m}) \cos 58.0^\circ \approx 38.366 \text{ m}, \\A_y &= (72.4 \text{ m}) \sin 58.0^\circ \approx 61.399 \text{ m}, \\B_x &= -(57.3 \text{ m}) \cos 36.0^\circ \approx -46.357 \text{ m}, \\B_y &= -(57.3 \text{ m}) \sin 36.0^\circ \approx -33.680 \text{ m}, \\C_x &= 0 \text{ m}, \quad C_y = -17.8 \text{ m}.\end{aligned}$$

3. Add components, keeping unrounded values.

$$R_x \approx -7.99 \text{ m}, \quad R_y \approx 9.92 \text{ m}, \quad \boxed{\mathbf{R} \approx (-7.99\hat{i} + 9.92\hat{j}) \text{ m}}.$$

The signs mean the keys are west and north of the starting point.

4. Find the direct distance.

$$R = \sqrt{R_x^2 + R_y^2} \approx \boxed{12.7 \text{ m}}.$$

5. Find the direction. The direction is in quadrant II. With $\beta = \tan^{-1}(|R_y/R_x|) \approx 51.1^\circ$,

$$\theta = 180^\circ - \beta \approx 128.9^\circ.$$

Equivalently, the direction is $\theta - 90^\circ \approx 38.9^\circ$ west of north. Report the direct route as

$$\boxed{12.7 \text{ m}, \quad 39^\circ \text{ west of north.}}$$

6. Check. The east–west and north–south components nearly cancel, so a short resultant is sensible. The full three-leg route is $72.4 + 57.3 + 17.8 = 147.5 \text{ m}$, much longer than the direct displacement’s magnitude.

Watch out

The value $\tan^{-1}(R_y/R_x) \approx -51.1^\circ$ lies in quadrant IV, while the components clearly place the resultant in quadrant II. These angles differ by 180° . Always check the component signs; the calculator cannot infer the intended quadrant from a single ratio.

1.10 The Scalar or Dot Product

Young–Freedman, Sec. 1.10, pp. 20–22.

Definition: A vector product whose result is a scalar

For nonzero vectors $\mathbf{A}$ and $\mathbf{B}$ separated by angle $0^\circ \leq \phi \leq 180^\circ$,

$$\boxed{\mathbf{A} \cdot \mathbf{B} = AB \cos \phi.}$$

It measures how strongly the vectors align. The dot product is a scalar; its units are the product of the units of the two vectors. It may be positive, zero, or negative:

Angle	Dot product	Interpretation
0°	AB	same direction
$0^\circ < \phi < 90^\circ$	positive	acute angle
90°	0	perpendicular
$90^\circ < \phi < 180^\circ$	negative	obtuse angle
180°	$-AB$	opposite directions

If either vector is zero, the dot product is zero although the angle is undefined. The converse “zero dot product implies perpendicular” therefore assumes both vectors are nonzero.

Derivation: The component formula

In perpendicular Cartesian axes,

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1, \quad \hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0.$$

Expand $(A_x \hat{i} + A_y \hat{j} + A_z \hat{k}) \cdot (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$. All cross-axis terms vanish, leaving

$$\boxed{\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z.}$$

For 2D vectors, omit the z -term. The dot product is commutative and distributive:

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}, \quad \mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}.$$

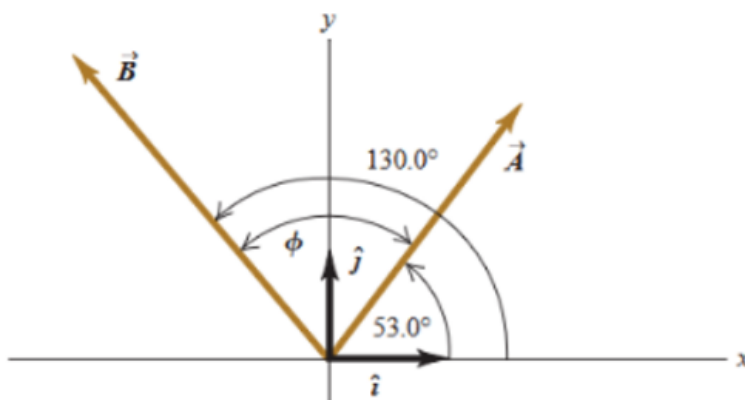
Also, $\mathbf{A} \cdot \mathbf{A} = A^2$, which is a useful magnitude check.

Example: The component example from the lecture

$$(3, 5) \cdot (4, 8) = 3 \times 4 + 5 \times 8 = 12 + 40 = \boxed{52}.$$

The answer is a single number, not $(12, 40)$. The vectors were given without physical units; if each had units of metres, the dot product would have units m^2 .

Figure 1.27 — Two routes to the same dot product

1.27 Two vectors in two dimensions.

Source: Young and Freedman, 13th ed., Fig. 1.27, p. 22; supplied figure image.

Read the angles. Vector $\mathbf{A}$ has magnitude 4.00 and direction 53.0° from $+x$; $\mathbf{B}$ has magnitude 5.00 and direction 130.0° . The angle *between* them is

$$\phi = 130.0^\circ - 53.0^\circ = 77.0^\circ.$$

Geometric calculation:

$$\mathbf{A} \cdot \mathbf{B} = (4.00)(5.00) \cos 77.0^\circ = 4.4990 \dots \approx \boxed{4.50}.$$

Component check:

$$\mathbf{A} = (4 \cos 53^\circ, 4 \sin 53^\circ) \approx (2.4073, 3.1945),$$

$$\mathbf{B} = (5 \cos 130^\circ, 5 \sin 130^\circ) \approx (-3.2139, 3.8302).$$

Then $A_x B_x + A_y B_y \approx 4.50$, as required. The horizontal contribution is negative and the vertical contribution positive; their sum is positive because 77° is acute. The two individual direction angles are not themselves the angle to insert into $AB \cos \phi$.

Example: Find the angle between two 3D vectors

The lecture gives

$$\mathbf{A} = 2\hat{i} + 3\hat{j} + \hat{k}, \quad \mathbf{B} = -4\hat{i} + 2\hat{j} - \hat{k}.$$

First compute the scalar product and magnitudes:

$$\mathbf{A} \cdot \mathbf{B} = 2(-4) + 3(2) + 1(-1) = -3, \quad A = \sqrt{14}, \quad B = \sqrt{21}.$$

Solve the geometric definition for the angle:

$$\phi = \cos^{-1} \left(\frac{\mathbf{A} \cdot \mathbf{B}}{AB} \right) = \cos^{-1} \left(\frac{-3}{\sqrt{14}\sqrt{21}} \right) \approx \boxed{100.1^\circ}.$$

A negative dot product requires an obtuse angle, which checks the result. This formula requires two nonzero vectors.

Example: Physical interpretation: work and projections

For a constant force and displacement,

$$W = \mathbf{F} \cdot \mathbf{d} = Fd \cos \phi.$$

Only the component of force along the displacement contributes to the work done by that force. If $F = 10 \text{ N}$, $d = 3 \text{ m}$, and $\phi = 60^\circ$, then $W = 15 \text{ J}$. A perpendicular force does zero work; a force component opposite to the displacement does negative work.

For a direction unit vector $\hat{\mathbf{n}}$, the signed scalar component of $\mathbf{A}$ along that direction is $\mathbf{A} \cdot \hat{\mathbf{n}}$. In particular, $A_x = \mathbf{A} \cdot \hat{\mathbf{i}} = A \cos \theta$.

1.11 The Vector or Cross Product

Young–Freedman, Sec. 1.10, pp. 23–25.

Definition: A vector perpendicular to the input vectors

In three-dimensional space, define $\mathbf{C} = \mathbf{A} \times \mathbf{B}$. For nonzero input vectors, its magnitude is

$$|\mathbf{C}| = AB \sin \phi, \quad 0^\circ \leq \phi \leq 180^\circ.$$

If the inputs are not parallel, $\mathbf{C}$ is perpendicular to both, with direction chosen by the right-hand rule. If the inputs are parallel, antiparallel, or one is zero, the cross product is the zero vector and has no direction.

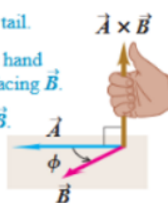
Its magnitude is the area of the parallelogram spanned by the two vectors: base A times perpendicular height $B \sin \phi$. When the input vectors are lengths, this area has units m^2 . In general, the cross product's units are the product of the input units.

Figure 1.29 — The right-hand rule and reversal of order

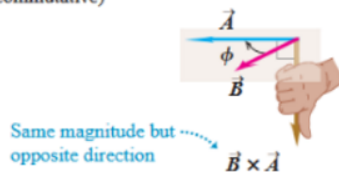
1.29 (a) The vector product $\vec{A} \times \vec{B}$ determined by the right-hand rule.
 (b) $\vec{B} \times \vec{A} = -\vec{A} \times \vec{B}$; the vector product is anticommutative.

(a) Using the right-hand rule to find the direction of $\vec{A} \times \vec{B}$

- ① Place $\vec{A}$ and $\vec{B}$ tail to tail.
- ② Point fingers of right hand along $\vec{A}$, with palm facing $\vec{B}$.
- ③ Curl fingers toward $\vec{B}$.
- ④ Thumb points in direction of $\vec{A} \times \vec{B}$.



(b) $\vec{B} \times \vec{A} = -\vec{A} \times \vec{B}$ (the vector product is anticommutative)



Source: Young and Freedman, 13th ed., Fig. 1.29, p. 23; supplied figure image.

Panel (a). Place the vectors tail to tail. Point your right-hand fingers along $\mathbf{A}$, then curl them toward $\mathbf{B}$ through the smaller angle. Your thumb gives the direction of $\mathbf{A} \times \mathbf{B}$, perpendicular to the plane of the two vectors.

Panel (b). Reversing the order reverses the direction but leaves the magnitude unchanged:

$$\mathbf{B} \times \mathbf{A} = -(\mathbf{A} \times \mathbf{B}).$$

The operation is therefore *anticommutative*. It is not enough to find $AB \sin \phi$: a complete nonzero cross product also needs a direction or signed components.

Theorem / Rule: Unit-vector products in a right-handed system

The forward cyclic order is $\hat{i} \rightarrow \hat{j} \rightarrow \hat{k} \rightarrow \hat{i}$:

$$\hat{i} \times \hat{j} = \hat{k}, \quad \hat{j} \times \hat{k} = \hat{i}, \quad \hat{k} \times \hat{i} = \hat{j}.$$

Reversing a product introduces a minus sign. A vector crossed with itself gives zero:

$$\hat{j} \times \hat{i} = -\hat{k}, \quad \hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \mathbf{0}.$$

For example, if $\mathbf{A} = 3\hat{i}$ and $\mathbf{B} = 4\hat{j}$, then $\mathbf{A} \times \mathbf{B} = 12\hat{k}$, while $\mathbf{B} \times \mathbf{A} = -12\hat{k}$.

Derivation: Component formula and determinant mnemonic

In a right-handed orthonormal Cartesian basis, a convenient arrangement is

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}.$$

Expand along the first row, using signs $+, -, +$:

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= (A_y B_z - A_z B_y) \hat{\mathbf{i}} \\ &\quad - (A_x B_z - A_z B_x) \hat{\mathbf{j}} \\ &\quad + (A_x B_y - A_y B_x) \hat{\mathbf{k}}. \end{aligned}$$

Equivalently, the three components are

$$\boxed{C_x = A_y B_z - A_z B_y, \quad C_y = A_z B_x - A_x B_z, \quad C_z = A_x B_y - A_y B_x.}$$

The minus sign in front of the $\hat{\mathbf{j}}$ term is already incorporated into the displayed formula for C_y ; do not apply it twice. You can use these component formulas without knowing general determinant theory.

Theorem / Rule: Useful checks and planar vectors

For $\mathbf{C} = \mathbf{A} \times \mathbf{B}$,

$$\mathbf{A} \cdot \mathbf{C} = 0, \quad \mathbf{B} \cdot \mathbf{C} = 0, \quad |\mathbf{C}| \leq AB.$$

If both input vectors lie in the xy -plane, append zero z -components:

$$(A_x, A_y, 0) \times (B_x, B_y, 0) = (0, 0, A_x B_y - A_y B_x).$$

With $+x$ right and $+y$ up on a page, positive z points out of the page ($\odot$), while negative z points into it ($\otimes$). The cross product is perpendicular to the page, not another arrow within it.

Watch out

The dot product uses **cosine** and produces a **scalar**; the magnitude of the cross product uses **sine** and the cross product produces a **vector**. For nonzero parallel vectors the cross product vanishes; for perpendicular vectors the dot product vanishes. Do not swap these statements.

Quick Reference and Common Exam Traps

Theorem / Rule: Core formula sheet

Speed and density: $v_{\text{av}} = d/\Delta t$, $\rho = m/V$,

Planar components: $A_x = A \cos \theta$, $A_y = A \sin \theta$,

Magnitude: $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$,

Sum components: $R_i = A_i + B_i$,

Dot product: $\mathbf{A} \cdot \mathbf{B} = AB \cos \phi$
 $= A_x B_x + A_y B_y + A_z B_z$,

Angle between vectors: $\cos \phi = \frac{\mathbf{A} \cdot \mathbf{B}}{AB}$,

Cross-product magnitude: $|\mathbf{A} \times \mathbf{B}| = AB \sin \phi$.

Here i in $R_i = A_i + B_i$ means any coordinate x, y, z ; it is not the unit vector $\hat{\mathbf{i}}$. The angle formula requires nonzero vectors.

$$1 \text{ N} = 1 \text{ kg m s}^{-2}, \quad 1 \text{ J} = 1 \text{ N m}, \quad 1 \text{ Pa} = 1 \text{ N m}^{-2},$$

$$1 \text{ N} = 10^5 \text{ dyne}, \quad 1 \text{ J} = 10^7 \text{ erg}, \quad 1 \text{ g cm}^{-3} = 10^3 \text{ kg m}^{-3}.$$

Feature	Dot product	Cross product
Result	scalar	vector
Geometric formula	$AB \cos \phi$	magnitude $AB \sin \phi$
Swap the inputs	unchanged	sign reversed
Parallel nonzero inputs	$+AB$ or $-AB$	zero vector
Perpendicular inputs	zero	magnitude AB
Physical application	work	torque

Watch out

Before accepting an answer, check these five things:

1. **Units:** have you converted grams to kilograms and squared or cubed the length conversion where needed?
2. **Angle convention:** is the given angle measured from $+x$, from north, or between two vectors?
3. **Signs and quadrant:** does the answer point in the direction indicated by the components?
4. **Type of answer:** should the result be a scalar, a vector, or a vector's magnitude?
5. **Size:** a resultant satisfies $|A - B| \leq |\mathbf{A} + \mathbf{B}| \leq A + B$, while $|\mathbf{A} \cdot \mathbf{B}| \leq AB$ and $|\mathbf{A} \times \mathbf{B}| \leq AB$.

Practice Problems

A. Multiple-choice questions

Choose one answer in each case.

1. A sample has density 1.2 g cm^{-3} . In SI units this is:
A 1.2 kg m^{-3} ; **B** 12 kg m^{-3} ; **C** 120 kg m^{-3} ; **D** 1200 kg m^{-3} ; **E** 12000 kg m^{-3} .
2. Which expression is equivalent to one joule?
A kg m s^{-1} ; **B** $\text{kg m}^2 \text{ s}^{-2}$; **C** $\text{kg m}^{-1} \text{ s}^{-2}$; **D** kg m s^{-2} ; **E** $\text{kg m}^2 \text{ s}^{-1}$.
3. A displacement is 10 m, 30° east of north. Its eastward component is:
A 5 m; **B** $5\sqrt{3}$ m; **C** 10 m; **D** -5 m; **E** 0 m.
4. Two nonzero vectors are perpendicular. Which statement must hold?
A Their sum is zero; **B** their magnitudes are equal; **C** their cross product is zero; **D** their dot product is negative; **E** their dot product is zero.
5. For $\mathbf{A} = 2\hat{\mathbf{i}}$ and $\mathbf{B} = 3\hat{\mathbf{j}}$, what is $\mathbf{B} \times \mathbf{A}$?
A $6\hat{\mathbf{k}}$; **B** -6 ; **C** $-6\hat{\mathbf{k}}$; **D** 0; **E** $6\hat{\mathbf{i}}$.
6. A vector has components $(-3, 4)$. Its magnitude and quadrant are:
A 7, II; **B** 5, II; **C** 5, IV; **D** -5 , II; **E** 1, I.
7. Convert 2.0 cm^3 to m^3 .
A 2.0×10^{-2} ; **B** 2.0×10^{-3} ; **C** 2.0×10^{-4} ; **D** 2.0×10^{-6} ; **E** 2.0×10^6 .
8. Vectors have magnitudes 3 and 4. Which cannot be the magnitude of their sum?
A 0; **B** 1; **C** 3; **D** 5; **E** 7.

B. Exercise 1.12: Vector operations

The vectors below are given without physical units. Use equal scales on the coordinate axes. In part (c), measure the angle from the positive x -axis in three dimensions. The labels $\mathbf{c}_{\text{sum}}$ and $\mathbf{c}_{\text{cross}}$ distinguish the results of parts (c) and (d).

- (a) Draw $\mathbf{u} = (2.5, 1.0)$ and $\mathbf{v} = (-0.5, -2.0)$. Plot $\mathbf{v} + \mathbf{u}$ and $\mathbf{u} - \mathbf{v}$. Calculate $\mathbf{u} \cdot \mathbf{v}$.
- (b) For $\mathbf{a} = (2, -4, 1)$ and $\mathbf{b} = (4, 5, 2)$, calculate $\mathbf{a} \cdot \mathbf{b}$.
- (c) Calculate $\mathbf{c}_{\text{sum}} = \mathbf{a} + \mathbf{b}$, its length, and its angle θ to the positive x -axis.
- (d) Calculate $\mathbf{c}_{\text{cross}} = \mathbf{a} \times \mathbf{b}$ and its length. Verify that your result is perpendicular to both inputs.

C. Displacement problems

9. **Fence posts.** A fence post is 52.0 m from where you are standing, in a direction 37.0° north of east. A second fence post is due south of you. What is the distance of the second post from you if the distance between the two posts is 80.0 m?
10. **A dog in an open field (Young–Freedman, Problem 1.83).** A dog runs 12.0 m east and then 28.0 m in a direction 50.0° west of north. In what direction and how far must the dog then run to end up 10.0 m south of her original starting point?

- 11. A sailboat in shifting winds (Young–Freedman, Problem 1.72).** A sailor sails 2.00 km east, then 3.50 km southeast, and then an additional distance in an unknown direction. Her final position is 5.80 km directly east of the starting point. Find the magnitude and direction of the third leg of the journey. Draw the vector-addition diagram and show that it is in qualitative agreement with your numerical solution.